

Web Applications Development Exam Assignment

June, 2026

Duration: 60 minutes

Additional Assignment

Extend your application with tag-based movie search. The table tags contains user-provided tags that characterize movie content. In the MovieLens small dataset there are 3684 tags, some of which are identical or very similar. Add a textbox and a button to the frontend. The user will type a search text and press the button. Although this functionality would normally be implemented using GET, the frontend must explicitly send an HTTP POST request to the backend.

Get movies for a tag

POST /tags/movies

Description: Returns all movies that have at least one tag matching the given keyword.

Request Body:

```
{
  "search": "keyword"
}
```

Response:

```
{
  "status": "success",
  "movies": [
    {
      "movieId": 296,
      "title": "...",
      "genres": "...",
      "matchingTag": "..."
    }, ...
  ]
}
```

Matching rule: - If the keyword has fewer than 5 characters, a tag matches only if it is exactly equal to the keyword. - If the keyword has at least 5 characters, a tag matches if the first 5 characters of the keyword are equal to the first 5 characters of the tag. - Matching must be case-insensitive.

The frontend must display the returned movies in a table.

Deliverables

Submit a compressed file named:

XXXX.zip

(where XXXXX are the last five digits of your student ID)

Frontend

A directory named "frontend" with the following files:

- `index.html`
- `index.js`
- `index.css`

Backend

A directory named "backend" with the following files:

- All source code files
- `requirements.txt` (or equivalent dependency manifest)
- `README.md` with setup and execution instructions
- Script(s) to create and populate the SQLite database
- Any configuration files required to run the backend

Miscellaneous

- You may use AI tools, but you must be able to explain your code.
- You must be able to demonstrate and extend your application.
- **Mark the new code that you added with comments: 10 dashes when it starts and 10 dashes when it finishes**
- **Late submissions will be penalized: -1 per minute counting from the first second.**